Linköping University
Institute for Analytical Sociology

Master's Thesis

Title

Descriptive Subtitle

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Abstract

The bachelor thesis examines the change in positive framing in sociological publications against the background of academic capitalism. Based on the theory of increasing self-promotion, it is analyzed whether this is reflected in scientific abstracts through increasingly positive language. The paper conducts a sentiment analysis of 121,789 English-language abstracts of sociological articles from the Web of Science database (1993-2023) with two dictionaries for additional validity. Interval plots, linear regressions and Monte Carlo simulations are used to examine temporal trends in positive and negative sentiment. The results show a very small, statistically insignificant increase in positive sentiment over time, while no clear trend can be identified for negative sentiment. The work reveals methodological challenges in the sentiment analysis of scientific texts, particularly with regard to data distribution and the significance interpretation of large samples. Contrary to the results of previous research, the study provides no evidence for a substantial increase in positive framing.

1 Acknowledgements

I would like to express my sincere gratitude to my supervisors, Prof. Dr. Elmar Schlüter and Tim Schmidt, for their invaluable guidance and support throughout this research project. Their insights and feedback have been instrumental in shaping this work.

I am also grateful to the Institute of Sociology at Justus Liebig University Giessen for providing the resources and environment conducive to academic research.

Finally, I would like to thank my family and friends for their unwavering support and encouragement during the completion of this thesis.

1.1 Generative AI Statement

This thesis was written with assistance from generative AI tools for specific purposes including literature research, text editing, and formatting. All substantive content, analysis, and conclusions are the result of my own intellectual work. AI-generated suggestions were critically evaluated and modified as appropriate. Complete transparency regarding AI usage has been maintained throughout the research process.

1.2 Ethics Statement

This research was conducted in accordance with the ethical guidelines of Justus Liebig University Giessen and the German Sociological Association. All data used in this study consists of publicly available academic publications. No human subjects were involved in this research, and therefore no ethics approval was required. The analysis was conducted with integrity, transparency, and respect for intellectual property rights.

2 Introduction

Question: How does the usage of populist rhetoric by german Members of Parliament on Twitter affect the politicians engagement communities?

H1: German MPs who use populist rhetoric have more strongly interconnected alters in their reply ego networks than those who do not.

- linguistic construction of in/outgroup
- hyping the ingroup –> selfselection effect keeps only engaged users with knowledge of groups social facts that actively reply to each other
- In results part:
  - In exploratory analysis, we operationalized alter disconnectedness as the fragmentation ratio (number of components divided by number of alters). Results mirror the main finding: lower fragmentation — i.e., more cohesive alter networks — is associated with populist rhetoric (OR = 0.02, p = .017).
  - Exploratory analysis using the fragmentation ratio (components / alters) corroborates this pattern. Populist MPs' reply networks are structurally more cohesive — their alters cluster into fewer, larger communities rather than responding in isolation (OR = 0.02, p = .017). This is consistent with the theoretical expectation that populist rhetoric fosters a sense of shared group identity among respondents, encouraging mutual engagement beyond dyadic replies to the politician.

H2: Do Populist MPs share more audience with each other than with non-populist MPs?

H3: What Network Structures are indicative of Populist MPs ego networks?

3 Literature Review

4 Data and Methods

Data were collected via Twitter's streaming API using two parallel strategies to capture reply interactions involving German MPs. The follow stream tracked replies to MP-generated tweets. The track stream captured tweets mentioning MP handles (e.g., "@username"), including direct replies and nested replies to reconstruct complete conversation threads. Protected accounts were excluded. Both datasets were combined and deduplicated to create the final reply dataset.

- Hat armin den Datensatz schonmal irgendwo verwendet, sodass ich ihn zitieren könnte?

1. First write this data collection section, send to armin for refinement and clarification purposes.
2. write a refined proposal that focuses on politicians as egonets

==> look in data where politicians are in reply network. make sure to filter first ofr ONLY the trees coming from politicians

5 Results

5.1 Text Analysis

Link Symetry of F1 score Appendix

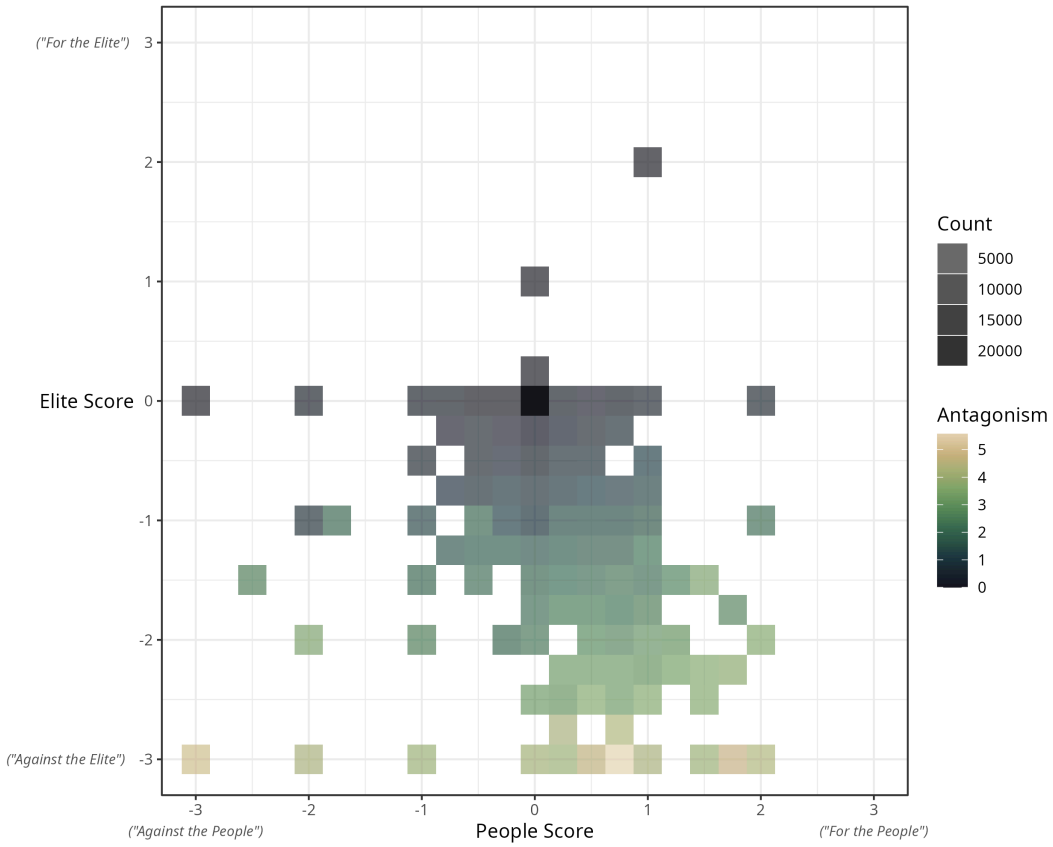

Figure 1: Title of the Figure

Here comes my long caption

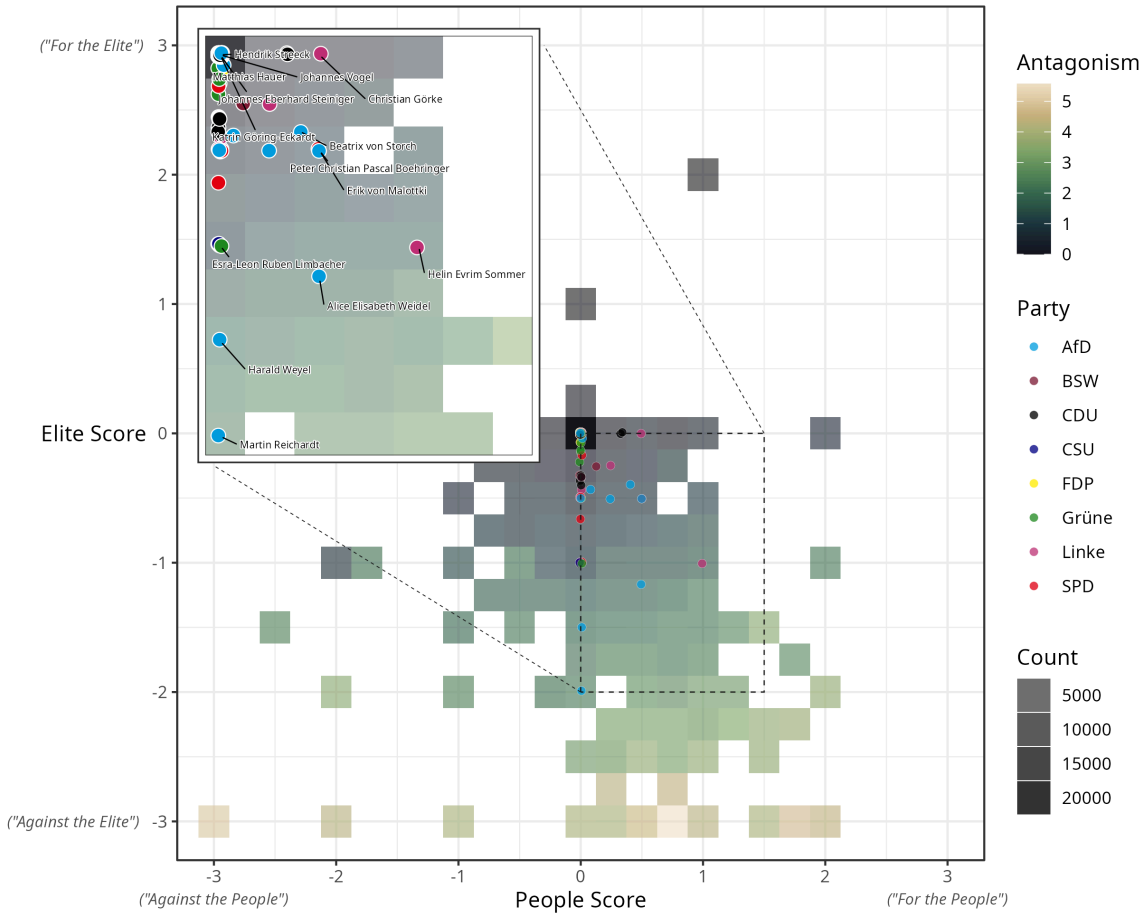

Figure 2: Title of the Figure

Here comes my long caption

5.2 Network Analysis

6 Discussion

7 Conclusion

7 Appendix A: Additional Tables, Figures and Equations

7 Symetry of F1 Score for swapping raters

Given binary ratings $a_i, b_i \in \{0, 1\}$ from two raters, define:

$$\text{both_positive} = \sum_i a_i b_i$$

$$\text{only}_{b_{\text{positive}}} = \sum_i (1 - a_i) b_i$$

$$\text{only}_{a_{\text{positive}}} = \sum_i a_i (1 - b_i)$$

Treating A as ground truth:

$$\text{precision} = \frac{\text{both_positive}}{\text{both_positive} + \text{only}_{b_{\text{positive}}}}, \quad \text{recall} = \frac{\text{both_positive}}{\text{both_positive} + \text{only}_{a_{\text{positive}}}}$$

Swapping roles swaps $\text{only}_{a_{\text{positive}}} \leftrightarrow \text{only}_{b_{\text{positive}}}$, which swaps precision and recall. Since F_1 is their harmonic mean:

$$F_1 = \frac{2 \cdot \text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} = \frac{2 \cdot \text{recall} \cdot \text{precision}}{\text{recall} + \text{precision}}$$

Multiplication and addition are commutative, so F_1 is invariant under the swap.

For swapping labels (0,1) F Scores are not symetric.

7 Appendix B: Code

```python
# Example: Data cleaning code
import pandas as pd
import numpy as np

# Load data
df = pd.read_csv('data.csv')

# Data processing steps
# ...
```

7 Appendix C: Prompts

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